Xin Tong

Image Scientist · Apple

xintong@caltech.edu | xintong-barry.github.io | Google Scholar

Education

2020 - 2026 **PhD, Medical and Electrical Engineering**

California Institute of Technology, Pasadena, CA, US

- GPA 4.0/4.0

2020 - 2021 **MS, Medical Engineering**

California Institute of Technology, Pasadena, CA, US

- GPA 4.0/4.0

2015 - 2019 **BS, Physics**

Peking University, Beijing, China

- GPA 3.8/4.0

Work Experiences

Present **Image Scientist**

Apple

2024/06 - 2024/09 **Research Scientist Intern, Advanced Display and Imaging**

Reality Labs Research, Meta

- Conducted research on advanced display technologies and imaging systems
- Focused on state-of-the-art hardware and computational methods

2020/01 - 2026 **Graduate Research Associate (Advisor: Prof. Lihong V. Wang)**

Caltech Optical Imaging Laboratory (COIL), Caltech

- Designed & optimized photoacoustic computed tomography for non-invasive vascular imaging
- Applied machine learning to reconstruct, enhance, and analyze images
- Developed quantum imaging systems with entangled-photon super-resolution and sub-shot-noise sensitivity

2018/07 - 2018/09 **Undergraduate Research Associate (Advisor: Prof. Aydogan Ozcan)**

Computational Imaging Lab, UCLA

- Implemented deep-learning algorithms for high-throughput holographic biosensor image analysis

2017/09 - 2018/07 **Undergraduate Research Associate (Advisor: Prof. Kebin Shi)**

Institute of Modern Optics, Peking University

- Researched wavefront-shaping techniques to achieve auto-focusing in optical systems

2017/07 - 2017/09 **Undergraduate Research Associate (Advisor: Prof. Dong Yu)**

Department of Physics, UC Davis

- Investigated spatially resolved photocurrent mapping using microscopy

Journal Articles

* Equal contributions. Publication years follow final journal issue metadata where available.

1. **Tong, X.***, Liu, C.Z.*, Luo, Y.*, Lin, L.*, Dzubnar, J., Invernizzi, M., Delos Santos, S., Zhang, Y., Cao, R., Hu, P., Zheng, J., Torres, J., Kasabyan, A., Lai, L.L., Yee, L.D., Wang, L.V.. Panoramic photoacoustic computed tomography with learning-based classification enhances breast lesion characterization. *Nature Biomedical Engineering*, 10(1), 161-177, 2026. doi:10.1038/s41551-025-01435-3

2. **Tong, X.***, He, Z.*, Zhang, Y.*, Liu, W.*, Huang, C.Y., Wang, L.V.. Above-twofold quantum super-resolution microscopy enabled by multiple idler passes with entangled biphotons. *Science Advances*, 12(30), eaea9457, 2026. doi:10.1126/sciadv.aea9457

3. Zhang, Y.*, Na, S.*, Russin, J.J.*, Sastry, K., Lin, L., Zheng, J., Luo, Y., **Tong, X.**, An, Y., Hu, P., Maslov, K., Tan, T.W., Liu, C.Y., Wang, L.V.. Rotational ultrasound and photoacoustic tomography of the human body. *Nature Biomedical Engineering*, 2026. doi:10.1038/s41551-025-01603-5

4. Delos Santos, S., Invernizzi, M., Liu, C., **Tong, X.**, Rosen, J., Wang, L.V., Lai, L.L.. Utilizing post-imaging surveys to guide development of photoacoustic computed technology for breast imaging. *The American Journal of Surgery*, 254, 116828, 2026. doi:10.1016/j.amjsurg.2026.116828

5. Hu, P., **Tong, X.**, Lin, L., Wang, L.V.. Data-Driven System Matrix Manipulation Enabling Fast Functional Imaging in Tomography. *IEEE Transactions on Medical Imaging*, 45(3), 906-918, 2026. doi:10.1109/TMI.2025.3612437

6. Luo, Y., Huang, H.K., Sastry, K., Hu, P., **Tong, X.**, Kuo, J., Aborahama, Y., Na, S., Villa, U., Anastasio, M.A., Wang, L.V.. Full-Wave Image Reconstruction in Transcranial Photoacoustic Computed Tomography Using a Finite Element Method. *IEEE transactions on medical imaging*, 44(2), 645-655, 2025. doi:10.1109/TMI.2024.3456595

7. Zhang, Y.*, He, Z.*, **Tong, X.***, Garrett, D.C., Cao, R., Wang, L.V.. Quantum imaging of biological organisms through spatial and polarization entanglement. *Science Advances*, 10(10), eadk1495, 2024. doi:10.1126/sciadv.adk1495

8. Zhang, Y.*, Hu, P.*, Li, L., Cao, R., Khadria, A., Maslov, K., **Tong, X.**, Zeng, Y., Jiang, L., Zhou, Q., et al.. Ultrafast longitudinal imaging of haemodynamics via single-shot volumetric photoacoustic tomography with a single-element detector. *Nature biomedical engineering*, 8(6), 712-725, 2024. doi:10.1038/s41551-023-01149-4

9. **Tong, X.**, Zhang, Y., Wang, L.V.. Enhancing Optical Microscopy with Quantum Entanglement. *Optics & Photonics News*, 2024. doi:10.1364/OPN.35.11.000032

Journal Articles (continued)

10. **Tong, X.***, Lin, L.*, Hu, P., Cao, R., Zhang, Y., Olick-Gibson, J., Wang, L.V.. Non-Invasive 3D Photoacoustic Tomography of Angiographic Anatomy and Hemodynamics of Fatty Livers in Rats. *Advanced Science*, 10(2), 2205759, 2023. doi:10.1002/advs.202205759

11. Lin, L.*, **Tong, X.***, Cavallero, S., Zhang, Y., Na, S., Cao, R., Hsiai, T.K., Wang, L.V.. Non-invasive photoacoustic computed tomography of rat heart anatomy and function. *Light: Science & Applications*, 12(1), 12, 2023. doi:10.1038/s41377-022-01053-7

12. He, Z.*, Zhang, Y.*, **Tong, X.***, Li, L., Wang, L.V.. Quantum microscopy of cells at the Heisenberg limit. *Nature Communications*, 14(1), 2441, 2023. doi:10.1038/s41467-023-38191-4

13. **Tong, X.***, He, Z.*, Zhang, Y.*, Solomon, S., Lin, L., Song, Q., Wang, L.V.. Experimental full-domain mapping of quantum correlation in Clauser-Horne-Shimony-Holt scenarios. *Physical Review Applied*, 19(3), 034049, 2023. doi:10.1103/PhysRevApplied.19.034049

14. Sastry, K., Zhang, Y., Hu, P., Luo, Y., **Tong, X.**, Na, S., Wang, L.V.. A method for the geometric calibration of ultrasound transducer arrays with arbitrary geometries. *Photoacoustics*, 32, 100520, 2023. doi:10.1016/j.pacs.2023.100520

15. Lin, L.*, Hu, P.*, **Tong, X.***, Na, S.*, Cao, R., Yuan, X., Garrett, D.C., Shi, J., Maslov, K., Wang, L.V.. High-speed three-dimensional photoacoustic computed tomography for preclinical research and clinical translation. *Nature Communications*, 12(1), 882, 2021. doi:10.1038/s41467-021-21232-1

16. Lin, L.*, **Tong, X.***, Hu, P., Invernizzi, M., Lai, L., Wang, L.V.. Photoacoustic computed tomography of breast cancer in response to neoadjuvant chemotherapy. *Advanced Science*, 8(7), 2003396, 2021. doi:10.1002/advs.202003396

17. Wu, Y., Ray, A., Wei, Q., Feizi, A., **Tong, X.**, Chen, E., Luo, Y., Ozcan, A.. Deep learning enables high-throughput analysis of particle-aggregation-based biosensors imaged using holography. *ACS Photonics*, 6(2), 294-301, 2018. doi:10.1021/acsphotonics.8b01479

18. Hou, Y., Xiao, R., **Tong, X.**, Dhuey, S., Yu, D.. In situ visualization of fast surface ion diffusion in vanadium dioxide nanowires. *Nano letters*, 17(12), 7702-7709, 2017. doi:10.1021/acs.nanolett.7b03832

Preprints

1. Cao, R.*, Luo, Y.*, Xu, J., Luo, X., Geng, K., Aborahama, Y., Cui, M., Davis, S., Na, S., **Tong, X.**, Liu, C., Sastry, K., Maslov, K., Hu, P., Zhang, Y., Lin, L., Zhang, Y., Wang, L.V.. Single-shot 3D photoacoustic computed tomography with a densely packed array for transcranial functional imaging. *arXiv preprint*, 2023. arXiv:2306.14471.

2. Zhang, Y., Na, S., Sastry, K., Russin, J.J., Hu, P., Lin, L., **Tong, X.**, Jann, K.B., Wang, D.J., Liu, C.Y., Wang, L.V.. Transcranial photoacoustic computed tomography of human brain function. *arXiv preprint*, 2022. arXiv:2206.00248.

Conference Publications and Abstracts

1. Yang, K., Park, S., Jeong, G., Liu, C.Z., **Tong, X.**, Cui, M., Ku, G., Villa, U.E., Wang, L.V., Anastasio, M.A.. Application of a learned SIR compensation method to improve elevational resolution in ring-array-based photoacoustic computed tomography. *Photons Plus Ultrasound: Imaging and Sensing 2026*, 13851, 138510J, 2026. doi:10.1117/12.3088475

2. **Tong, X.**, Liu, C.Z., Luo, Y., Lin, L., Dzubnar, J., Invernizzi, M., Zhang, Y., Cao, R., Hu, P., Torres, J., Kasabyan, A., Lai, L.L., Yee, L.D., Wang, L.V.. Integrating machine learning with panoramic photoacoustic computed tomography for improved breast lesion analysis (Conference Presentation). *Photons Plus Ultrasound: Imaging and Sensing 2025*, 2025. doi:10.1117/12.3047914

3. Lin, L., Hu, P., **Tong, X.**, Na, S., Cao, R., Yuan, X., Garrett, D.C., Shi, J., Maslov, K., Wang, L.V.. Three-dimensional photoacoustic computed tomography for preclinical research and clinical translation. *Photons Plus Ultrasound: Imaging and Sensing*, 2021.

4. Lin, L., **Tong, X.**, Hu, P., Invernizzi, M., Lai, L.L., Wang, L.V.. Clinical photoacoustic computed tomography of breast cancer treated with neoadjuvant chemotherapy. *Photons Plus Ultrasound: Imaging and Sensing*, 2021.

5. Wu, Y., Ray, A., Wei, Q., Feizi, A., **Tong, X.**, Chen, E., Luo, Y., Ozcan, A.. Particle-Aggregation Based Virus Sensor Using Deep Learning and Lensless Digital Holography. *CLEO: Applications and Technology*, 2019.

6. Hou, Y., Xiao, R., **Tong, X.**, Dhuey, S., Yu, D.. Direct Visualization of Fast Surface Ion Diffusion in Vanadium Dioxide Nanowires. *APS March Meeting Abstracts*, 2018.

Honors and Awards

2024 · Winning Place (3rd Place), International Photography Awards; Honorable Mention, International Photography Awards

2019 · National Scholarship, Peking University; Merit Student Pacesetter, Peking University; Outstanding Graduates, Peking University

2018 · CSST Scholarship, UCLA

Professional Activities

Conference Oral Presentations

- Photoacoustic and quantum enhanced imaging (invited talk). *22nd Annual Neurotech Convention*, SBMT, 2025.

- Integrating machine learning with panoramic photoacoustic computed tomography for improved breast lesion analysis. *In Photons Plus Ultrasound: Imaging and Sensing*. SPIE Photonics West, 2025.

- Super-resolution quantum microscopy at the Heisenberg limit. *In Quantum Sensing, Imaging, and Precision Metrology II*. SPIE Quantum/Photonics West, 2024.

- Experimental full-domain mapping of quantum correlation in Clauser-Horne-Shimony-Holt scenarios, *In Quantum Computing, Communication, and Simulation IV*. SPIE Quantum/Photonics West, 2024.

- Non-invasive photoacoustic computed tomography of cardiac anatomy and function in rats. *In Photons Plus Ultrasound: Imaging and Sensing*. SPIE Photonics West, 2023.

- Three-dimensional photoacoustic tomography of angiographic anatomy and hemodynamics for fatty liver study. *In Photons Plus Ultrasound: Imaging and Sensing*. SPIE Photonics West, 2023.

Teaching Activities

- Teaching assistant, MedE 168: Biomedical Optics: Principles and Imaging, Caltech

- Teaching assistant, MedE/EE 268: Medical Imaging, Caltech

- Teaching assistant, APh/Ph 138b: Quantum Hardware and Techniques, Caltech

Peer Review Services

- 100+ peer review records in *Light: Science & Applications*, *IEEE Transactions on Medical Imaging*, *Photoacoustics*, etc.

Guest Editor Services

- Technical Program Committee Member, special issue *Advanced Optical Detection and Imaging Systems*, *Electronics*.

Interests and Skills

Research Interests

- Optical Imaging (e.g., Quantum Imaging, Photoacoustic Imaging, Computational Imaging)
- Image Processing and Reconstruction
- Medical Devices and Translational Research
- Machine and Statistical Learning
- Quantum Optics and Photonics

Technical Skills

- **Programming languages and APIs**: MATLAB, Python, C/C++, LaTeX; Numpy, TensorFlow, XGBoost, etc.

- **Software**: LabVIEW, Zemax OpticsStudio, SolidWorks, ImageJ, Adobe Photoshop, Lightroom, Premiere Pro, etc.

Hobbies

- Travel and Portrait Photography
- Hiking
- Skiing